ing of the water tap by different people can spread the germs. So, this touchless water tap will take care of that as well.

Flow chart of this low-cost, easy-to-make water dispenser for washing hands is shown in Fig. 1. The components required to make it are mentioned under the Bill of Material table. Circuit diagram for the touchless water dispenser is shown in Fig. 2.

The circuit comprises an Arduino board (MOD1), ultrasonic sensor HC-SR04 (MOD2), transistor BC547 (T1), 5V one changeover relay, a cooler water pump, and a few other components listed under the Bill of Material table. The relay energises automatically when a user's hands are just 3 to 5cm away from the ultrasonic sensor to deliver water.

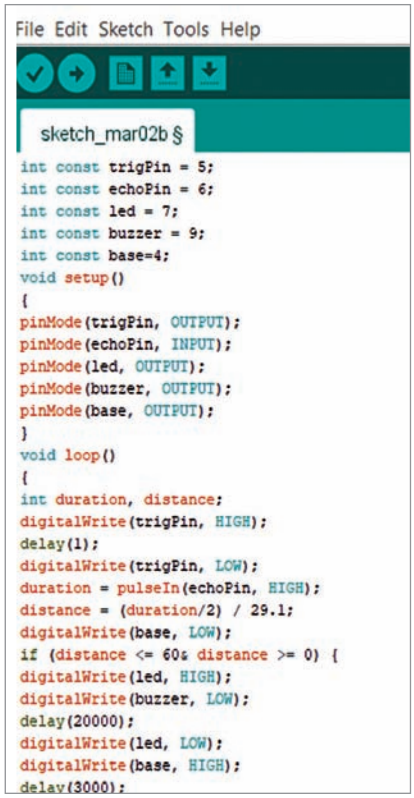
Install Arduino IDE in the Arduino board and upload the source code (see Fig. 3 for source code snippet) by selecting the right board and port number in Arduino IDE.

Assembling and testing

After uploading the source code, assemble the circuit on a general-purpose PCB and enclose it in a suitable box. Install the ultrasonic sensor in front of the wash basin in such a way that when someone brings hands near it, the tap water starts flowing.

After proper installation, power on the circuit. When someone who wants to wash his/her hands comes in front of the ultrasonic sensor, within 60cm, the LED automatically lights up. It indicates that the person should start rubbing hands with soap.

After 20 seconds, the relay automatically drives the pump motor for 3 seconds to dispense water for washing hands. Then the pump mo-



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sketch_mar02b §

```
int const trigPin = 5;
int const echoPin = 6;
int const led = 7;
int const buzzer = 9;
int const base=4;
void setup()
{
pinMode(trigPin, OUTPUT);
pinMode(echoPin, INPUT);
pinMode(led, OUTPUT);
pinMode(buzzer, OUTPUT);
pinMode(base, OUTPUT);
}
void loop()
{
int duration, distance;
digitalWrite(trigPin, HIGH);
delay(1);
digitalWrite(trigPin, LOW);
duration = pulseIn(echoPin, HIGH);
distance = (duration/2) / 29.1;
digitalWrite(base, LOW);
if (distance <= 60& distance >= 0) {
digitalWrite(led, HIGH);
digitalWrite(buzzer, LOW);
delay(20000);
digitalWrite(led, LOW);
digitalWrite(base, HIGH);
delay(3000);
```

Fig. 3: Code snippet

tor stops for 2 seconds so the person can check her/his hands. The motor starts delivering water once again after 2 seconds for another 3 seconds to ensure the hands have been cleaned thoroughly. Thereafter, the buzzer rings for 3 seconds to indicate that hands washing process has been completed and now the person may remove hands from the washbasin and leave the place. EFY

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EFY Note
The source code of this project is available for free download at *www.electronicsforu.com*